Title

Computer Space

Handheld Controller for Test Environments

(Prerelease Version)

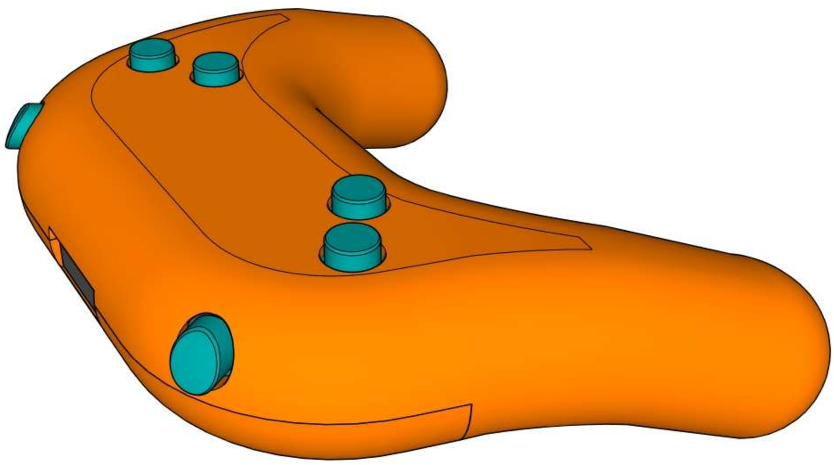


Table of Content

Preface

Some handheld controller for such a unique and beautiful cabinet like Computer Space?
Does this make sense?

Well not for normal operation. I liked to have something small and usable for the tight space of my electronics lab as it is quite impractical to use the original control panel for the purpose of maintaining the electronic boards.

For some time, I used a quick and dirty soldered prototype PCB with a bunch of microswitches. While it worked, I always felt that it might require some improvement.

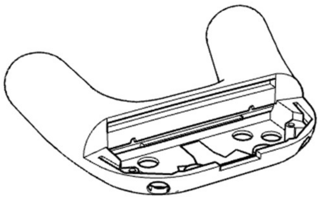
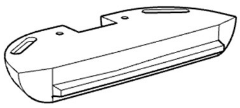
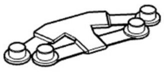
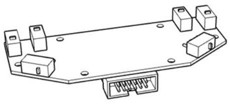
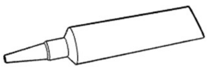
To give the involved efforts more sense, the design files together with this manual are shared with the community.

Please be aware, that the 12 pin IDC connector is pin-compatible with the CS backplane PCB that was released earlier on the Github repository.

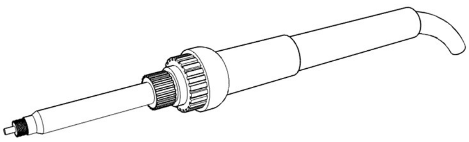
Have fun with the build and happy arcade gaming,
Stefan Burger

Materials and Tools

Materials

ID	Illustration	Description	Quantity
M1		CS-controller_top.stl	1
M2		CS-controller_bottom.stl	1
M3		CS-controller_action.stl	1
M4		CS-controller_shoulder-left.stl	1
M5		CS-controller_shoulder-right.stl	1
M6		Custom PCB	1
M7		M3x12 screws	2
M8		M3x10 screws	2
M9		M3 threated inserts	2
M10		Super Glue	1

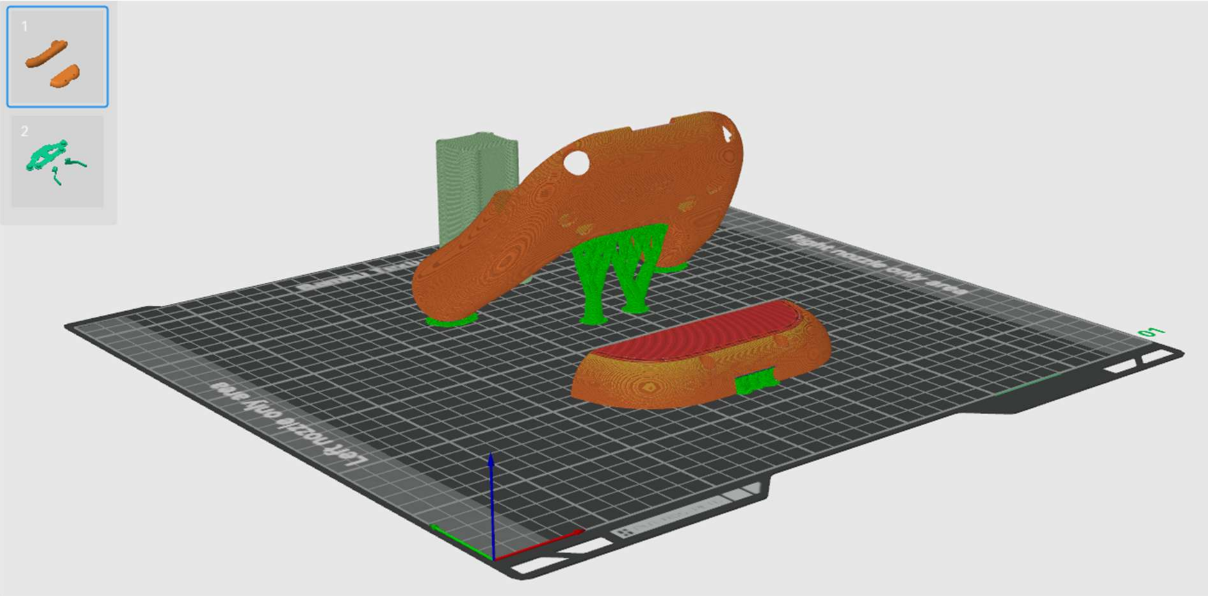
Tools

ID	Illustration	Description
T1		M3 Allen Key
T2		Threated Insert application Tool

3D Printing

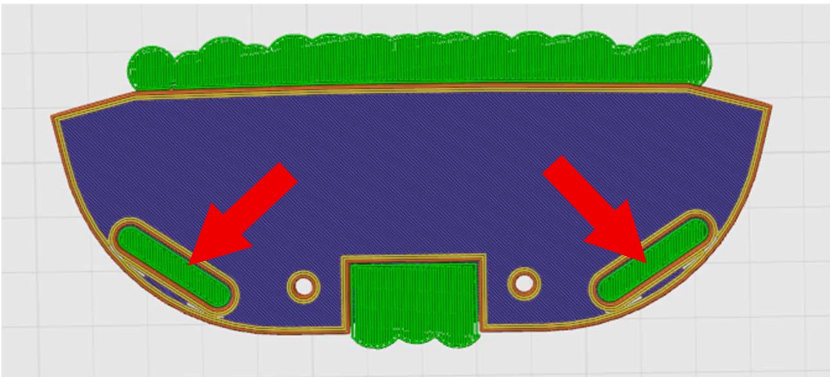
In general, all 5 models should fit on a typical 256 x 256 build plate. To go with a multicolored scheme, either use the multi-material capabilities of your printer or print housing parts and button inlays in two steps on separate plates.

Housing



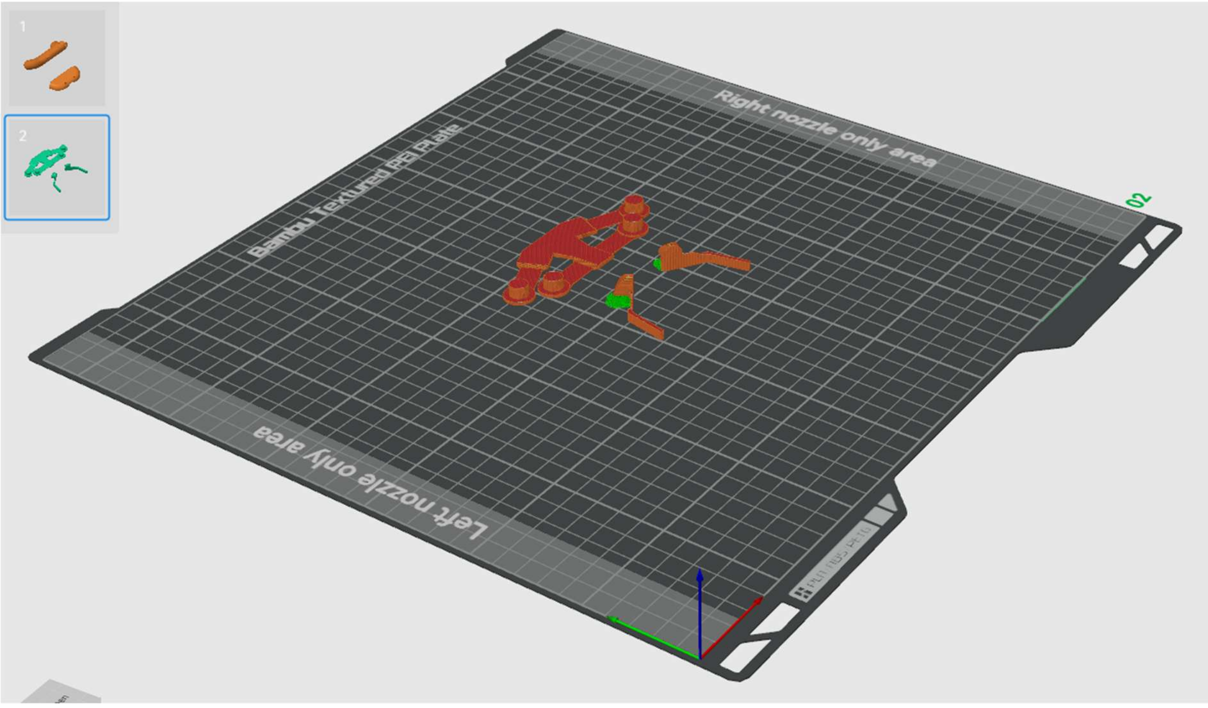
As the internal geometry of the housings top-side is quite delicate and every tenth of a millimeter for the tolerances can be quite critical, the print orientation was optimized to prevent any artifacts of support material or other artefacts. The best possible print orientation for the top-side element are 122 degrees, as illustrated in the picture above.

The bottom side of the housing has two diagonal recesses that are necessary to prevent friction of the shoulder buttons flexing tab with the housings bottom part. Print supports can create a visually closed surface and the removal of those supports can easily be overseen (ask how I know it). So, it might be worth to double check their proper removal before assembly begins.



Besides of that, the two parts were printed in PLA+ with 3 outer walls and 10% infill.

Button Inserts



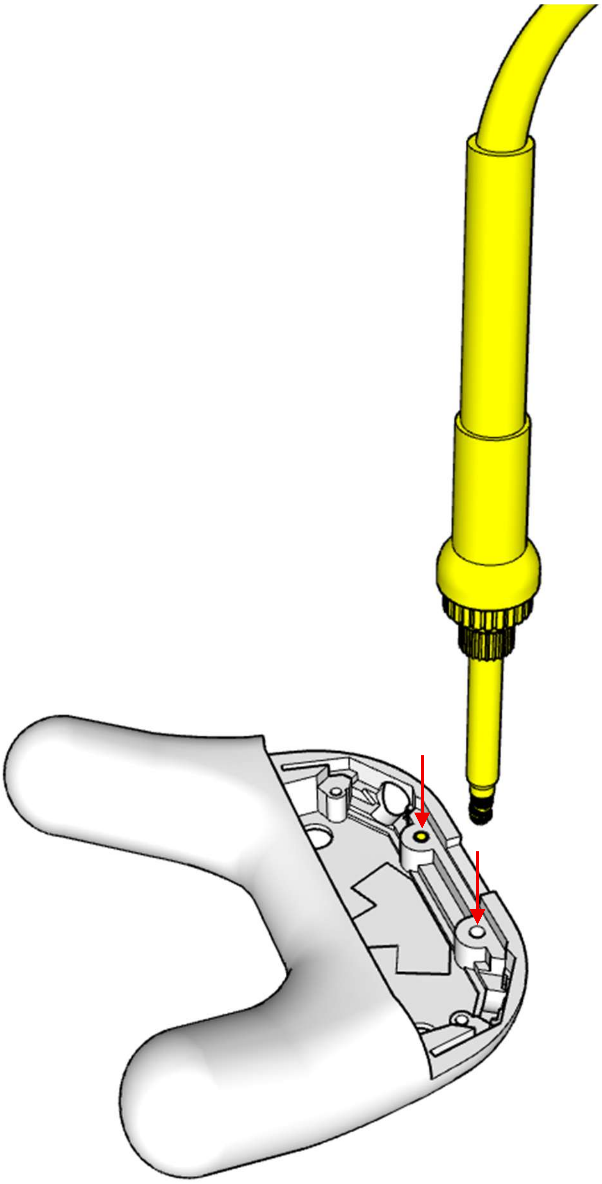
All button inlays are designed with the button itself, attached to a flexing tab and an anchor part that gets attached to the housing (all in one piece). Because the connecting tab has to bend without breaking, the illustrated print orientation is mandatory to prevent breaking along layer lines while bending.

All button inlays are printed with PLA+, 3 outer walls and 100% infill (all together, less then 10 grams of filament).

Assembly

Step 1 – Apply M3 threated inserts

Use a soldering iron or a specific tool for applying the two threated inserts into the two front holes of the upper housing part. Be aware that in case a soldering iron is used with a very thin and long tip, it might penetrate unintentionally the outside of the housing.

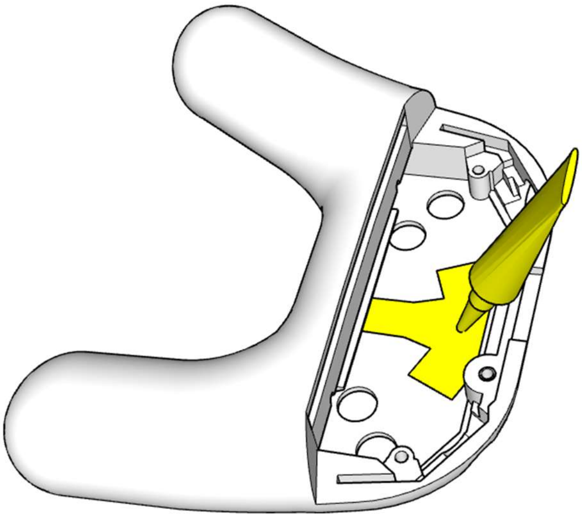


Materials: M1, M9 (x2)

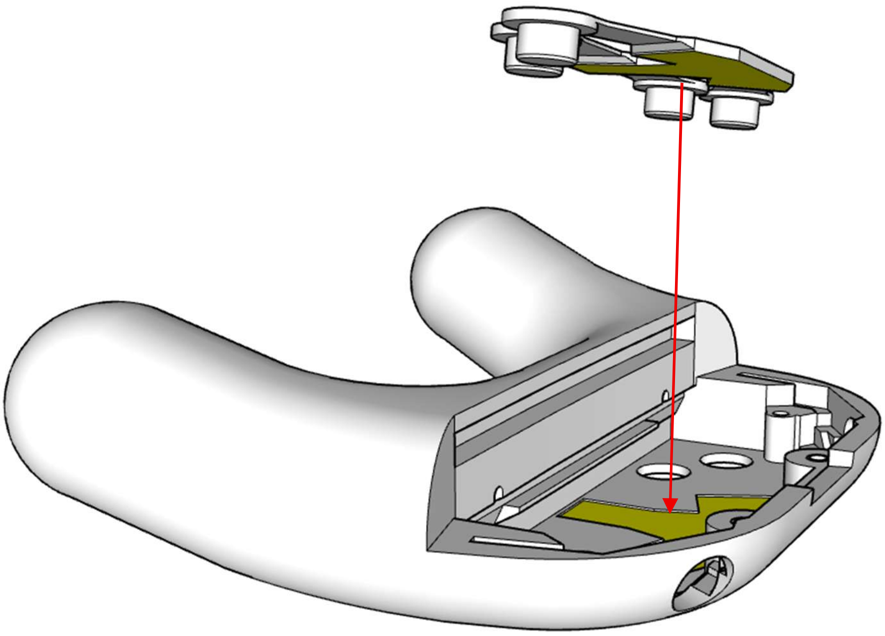
Tool: T2

Step 2 – Glue Action Button inlay into Housings Top

Apply a reasonable and moderate amount of super glue to the T-shaped recess. The recess is not only an indication of the relevant area; it also should prevent the glue from overfloating other areas (due to surface tension) and also makes the alignment of the action button inlay with the housing top much easier.



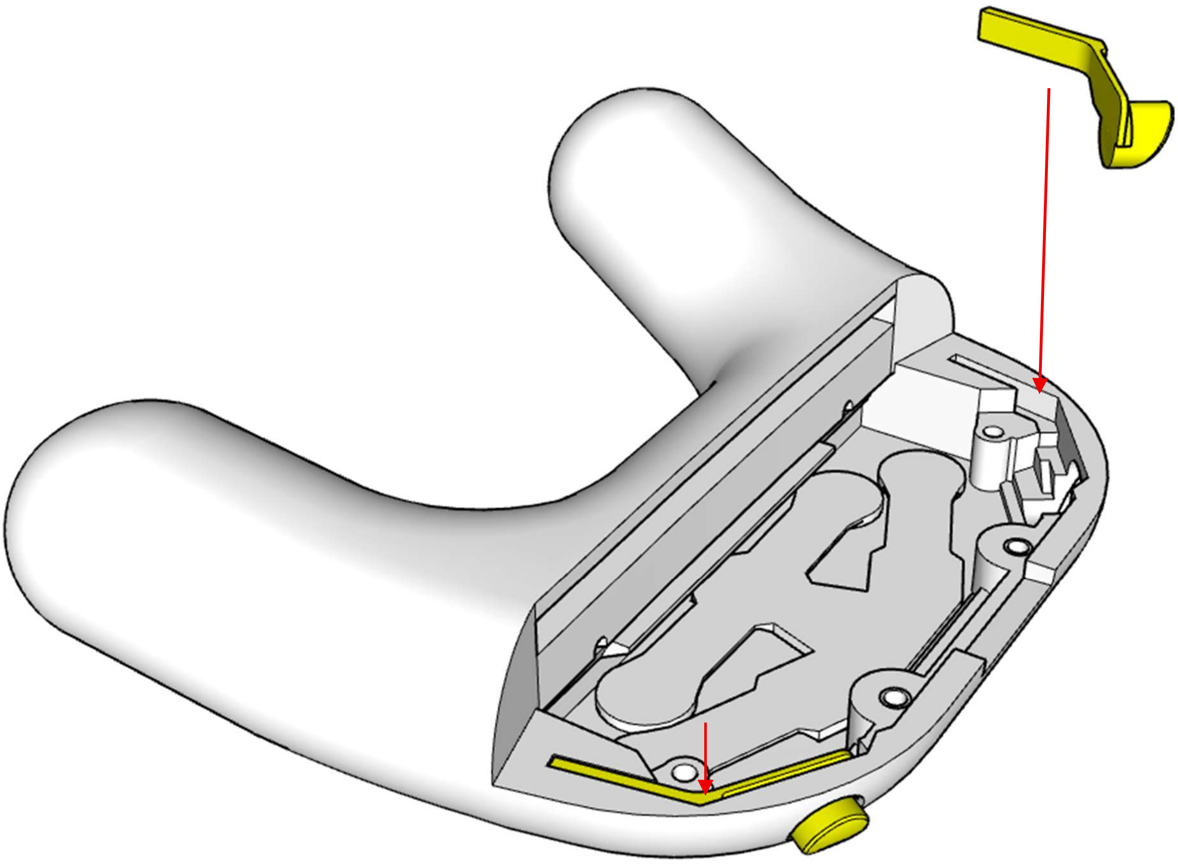
Materials: M1, M10



Materials: M1, M3

Step 3 – Insert Shoulder Buttons into Housings Top

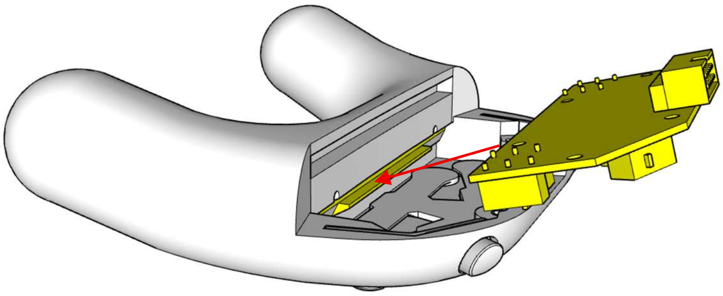
Apply the two shoulder buttons to the button holes and grooves on the outer left and right side. Go with the button through the button hole in a first step, then push the long straight anchor into the groove. When correctly applied, the buttons curved front should follow the curved housing outer side and the top side of the straight anchor should sit flat with the housing.



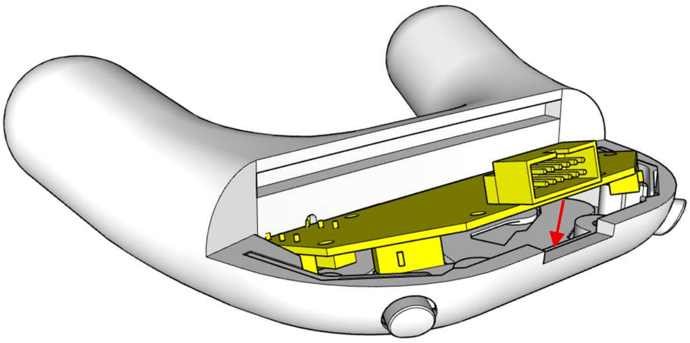
Materials: M1, M4, M5

Step 5 – Apply Custom PCB to Housings Top

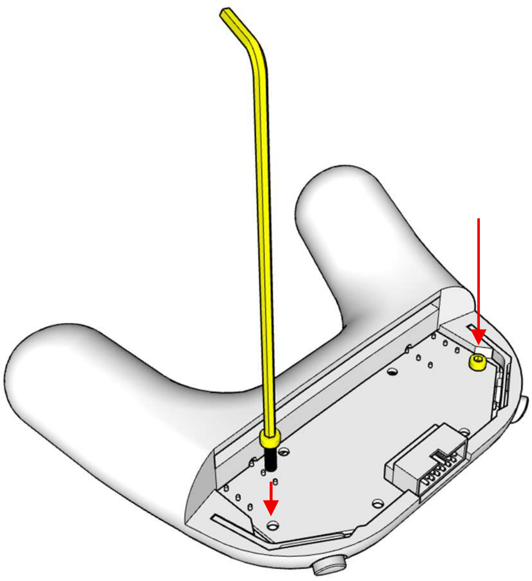
Orient the PCB with the IDC connector pointing up and the diagonals pointing to the front as indicated in the illustration. Align the long straight edge of the PCB with the groove in the back of the housing.



As soon as that edge sits inside the groove, lower the front side until it sits flat inside the housing.



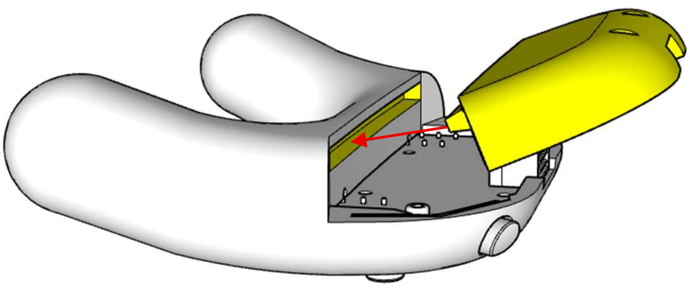
Finaly, secure PCB with two M3x10 screws on the left and right side to the top of the housing



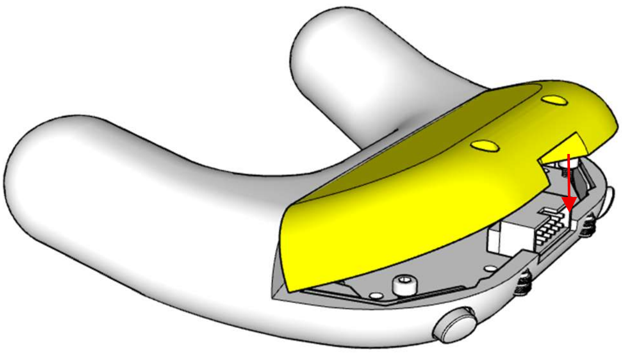
Materials: M1, M8 (x2); Tool: T1

Step 6 – Apply Housing Bottom to Housings Top

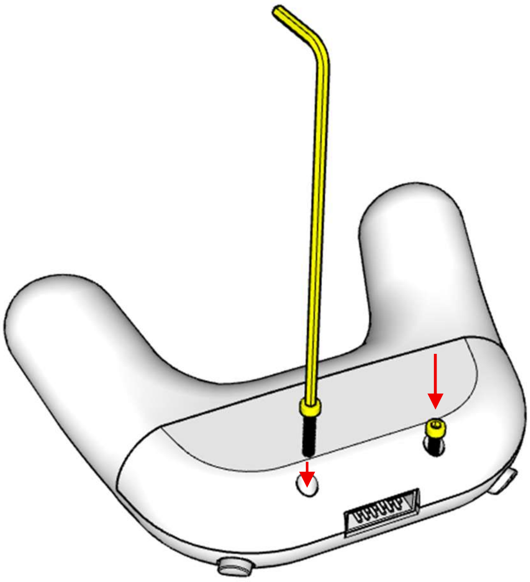
Align the lip of the housings bottom with the groove at the back of the housings top as illustrated.



Once the bottom housing is completely moved to the back of the housings top, lower the front of the bottom housing until it sits flat with the top housing.



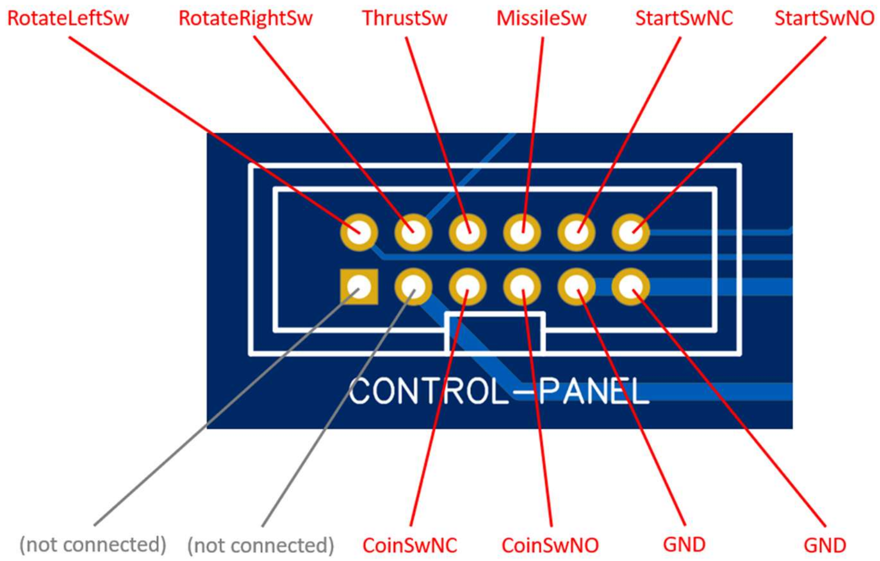
Finally secure the housing bottom to the housing top with x2 M3x12 screws.



Materials: M1, M2, M7
Tool: T1

Connectivity

The pin layout of the IDC connector is as follows:



It is compatible with the CS backplane PCB that was released on Github earlier: